INCH-POUND
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PERFORMANCE SPECIFICATION

SEMITRAILER, VAN: 10 TON, 4-WHEEL SERIES

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1. Scope. This specification covers military-designed semitrailers, van, 10 ton, 4-wheel to be used for housing electronic and similar sensitive equipment.

2. Type classification (see table I).

TABLE I. Classification (see 6.2).

Vehicle model	Army part number	Vehicle type
XM557	8736376	Semitrailer, Van: Electronic
XM558	8736377	Semitrailer, Van: Electronic
XM574	8736436	Semitrailer, Van: Electronic
MX574E1	8736767	Semitrailer, Van: Electronic
MX574E2	8736926	Semitrailer, Van: Electronic
XM654	8736438	Semitrailer, Van: Electronic Telemetering
XM680	8736606	Semitrailer, Van
XM680E1	8736689	Semitrailer, Van

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2330

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TABLE I. Classification (see 6.2) - Continued.

Vehicle model	Army part number	Vehicle type
XM703	8736699	Semitrailer, Van
XM738	8736752	Semitrailer, Van: Telephone
XM739	8736754	Semitrailer, Van: Switchboard
XM739E1	8736879	Semitrailer, Van: Switchboard
XM822	8736902	Semitrailer, Van: Petroleum Laboratory
XM823	8736903	Semitrailer, Van: Teletype
XM824	8736904	Semitrailer, Van: Crypto
XM844	8736932	Semitrailer, Van: On-line
XM845	8736933	Semitrailer, Van: Off-line
XM847	8736934	Semitrailer, Van: Dig Term 1
XM848	8736935	Semitrailer, Van: Dig Term 2
XM849	8736936	Semitrailer, Van: Sec Voice
XM850	8736937	Semitrailer, Van: Voice Switch
XM703E1	8736951	Semitrailer, Van: Elect Equip
XM912	8736969	Semitrailer, Van: Main Frame
XM913	8736970	Semitrailer, Van: Mass Storage
XM971	8736998	Semitrailer, Van: Data Center
XM971E1	8750087	Semitrailer, Van: Data Center
XM971E2	8750090	Semitrailer, Van: Satellite Terminal Operation
XM971E3	8750091	Semitrailer, Van: Satellite Terminal, Maintenance & Supply
XM991	8750048	Semitrailer, Van: Repair Facility
XM991E1	8750092	Semitrailer, Van: Repair Facility
XM991E2	8750096	Semitrailer, Van: Central
XM995	8750049	Semitrailer, Van: Test Facility
XM995E1	8750093	Semitrailer, Van: Test Station
XM822E1	8750094	Semitrailer, Van: Petroleum Laboratory
XM995E2	8750097	Semitrailer, Van: Mass Storage

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to

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ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurements.

DEPARTMENT OF DEFENSE

MIL-STD-210	- Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment.
MIL-STD-285	- Attenuation Measurements for Enclosures, Electromagnetic Shielding, for Electronic Test Purposes, Method of
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-642	- Identification Marking of Combat, and Tactical Transport Vehicles.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations (FMCSR)
Federal Motor Vehicle Safety Standards (FMVSS)

(Copies of the Department of Transportation publications are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

THE TIRE AND RIM ASSOCIATION, INC.

Year Book.

(Application for copies of The Tire and Rim Association publications may be obtained from The Tire and Rim Association, Inc., 175 Montrose West Ave., Suite 150, Copley, OH 44321.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for semitrailers. The materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.6.1).

3.3 Ozone resistant. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Construction. The semitrailer shall be constructed in accordance with the drawings and specifications applicable to the specific vehicle on contract. If more than one unit of a particular component is used in or on the semitrailer, the components used shall be identical in make, material, and quality. The riveting, welding practices and quality shall be the same on each semitrailer (see 4.6.1 and 4.6.2).

3.4.1 Spare wheel assembly. One spare wheel, tire, and tube assembly shall be furnished with each semitrailer (see 4.6.3).

3.4.1.1 Tires. Unless otherwise specified (see 6.2), tires shall be cross-country in accordance with the Tire and Rim Association.

3.4.1.2 Inner tubes. Unless otherwise specified (see 6.2), inner tubes shall be of the heavy duty type and shall be of proper size for tires furnished. Tube type radial tires shall have radial type inner tubes. Tire flaps shall be provided for tube type tires in accordance with the Tire and Rim Association recommendations.

3.4.2 Electrical system. Electrical system shall be 24 volts direct current (dc) (nominal) (see 4.6.4).

3.4.2.1 Lighting system. The internal and external lighting systems as specified on applicable drawings, shall operate on voltage between 18 and 30 volts under all semitrailer operating conditions (see 4.6.4).

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3.4.2.2 Electrical circuits. The electrical circuits shall maintain continuity from end to end without any shorts from pin to pin and from pin to connector shell. All electrical cables and wiring shall be anchored in such a manner to prevent any damage to cables and wiring during all semitrailer operating conditions (see 4.6.4).

3.4.3 Body air leakage. The van body shall be so constructed that when all doors and openings are closed, the body shall be airtight to the extent that air pressure of not less than 0.5 inch (in.) of water column pressure will be developed within the semitrailer when air at 100 cubic feet per minute (ft^3/min) and at 50 pounds per square inch (psi) is forced from outside of van body to the inside (see 4.6.5).

3.4.4 Waterproofing and sealing. The van body shall be capable of protecting the equipment installed in its interior from rain blowing at the rate of 4 in. per hour with water droplets averaging 0.1 in. diameter (minimum 0.06 in.) and impinging upon the vehicle at any angle between 0 degrees and 45 degrees from the vertical for 30 minutes. No rain water is permitted to enter the interior of the van body. Openings required to be open during operation of the interior equipment (e.g., air conditioners) must be provided with covers and side flaps preventing rain at a maximum angle of 45 degrees from the vertical entering the van (see 4.6.6).

3.4.5 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment settings during all semitrailer operating conditions (see 4.6.7).

3.4.6 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all semitrailer operating conditions (see 4.6.8).

3.4.7 Air, hydraulic lines and fittings. Provisions shall be made to assure internal cleanliness of air lines, hydraulic lines, and fittings prior to connection and initial use of semitrailer. All air and hydraulic lines and connections shall be free of leaks. All air and hydraulic lines and fittings shall be anchored in such a manner to prevent any damage during all semitrailer operating conditions (see 4.6.9).

3.4.8 Lifting attachments. The semitrailer shall be provided with lifting attachments which shall be structurally sound to lift the semitrailer with rated payload (see 4.6.10).

NOTE: Spreader bar used for lifting; spreader bar not furnished with vehicle.

3.4.9 Tiedown devices. The semitrailer shall be provided with tiedown devices sufficient in number and structurally sound for transporting by rail, sea, and airborne operations (see 4.6.10).

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3.4.10 Landing gear and leveling jacks. The landing gear, when supporting the front part of the semitrailer while the rear part of the vehicle is resting on the suspension, shall withstand without damage the strains imposed when coupling and uncoupling the semitrailer to and from the towing vehicle. Landing gear and leveling jacks shall be capable of supporting the semitrailer and the evenly distributed payload and they shall operate freely when the semitrailer is lifted and lowered. Swing-up type landing gear shall be freely rotatable into the horizontal position during aircraft loading and unloading, when specified (see 4.6.11).

3.4.11 Seals. When fording or operating in mud, sand, or snow, the seals shall be capable of preventing the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearing seals shall restrict the leaking of lubricants from bearings (see 4.6.12).

3.4.12 Safety. The trailer shall comply with all Federal Motor Carrier Safety Regulations (FMCSR) and Federal Motor Vehicle Safety Standards (FMVSS) requirements applicable to this vehicle at the time of manufacture. The semitrailer shall be free of sharp projections or edges that may cause the operator or maintainer injury. All rotating or moving parts, hot surfaces, electrically energized or high pressure components, or other inherently hazardous items shall be effectively guarded or insulated to protect operating and maintenance personnel.

3.5 Performance. Each semitrailer shall meet performance requirements specified herein with basic vehicle equipment as depicted on the drawings and serviced with products listed in table V. Semitrailer shall transport actual rated payload or simulated load of equal weight and distribution when towed by a truck tractor, 5 ton, 6x6, M52 (see 4.6.13).

3.5.1 Turning ability. The loaded semitrailer coupled to the towing vehicle or dolly shall make right and left turns within its minimum turning circle. The semitrailer shall assume an angle of 90 degrees to the towing vehicle without damage to the loaded semitrailer and towing vehicle, and without interference between the semitrailer and the towing vehicle (see 4.6.14).

3.5.2 Service brakes. The service brakes of the loaded semitrailer and towing vehicle shall control, decelerate, and stop the combination with the rated payload within 30 feet from a speed of 20 mph on dry, hard, relatively level and smooth surface road. Hydraulic fluid and air leakage shall not be permitted. Application of brakes on all wheels of the semitrailer and towing vehicle shall be simultaneous (see 4.6.15).

3.5.2.1 Automatic actuation. The automatic brake-actuation device of the semitrailer shall automatically actuate the service brakes to stop and hold the semitrailer upon breakaway from the towing vehicle. The service brakes shall then hold the loaded semitrailer stationary when on grades up to and including 20 percent for a period of not less than 15 minutes (see 4.6.15).

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3.5.3 Fording. The semitrailer, without the use of special kits, shall ford fresh or salt water crossing 60 in. deep for a period of 15 minutes without water leaking into the interior of the van body and damaging the submerged components, such as suspension, air-mounted kingpin, brake system, wheels, jacks, and electrical components, including ancillary equipment to the trailer mounted or stowed beneath the water level. Other equipment, such as generators and gasoline engines, a submerged during fording, shall be protected by special covers or kits as specified by the contractual requirements (see 4.6.16).

3.5.4 Environmental. The semitrailer, when properly serviced, shall operate and transport equipment or cargo under any extreme condition of inclement weather and in ambient air temperatures from -50 degrees Fahrenheit (°F) to +125°F, without indication of deterioration. When in storage, the complete semitrailer shall withstand climatic extremes, as specified in MIIL-STD-210, without deterioration that may cause failure of any component of the semitrailer (see 4.6.17).

3.5.5 Towing speeds and trailing ability. The semitrailer, attached to the towing vehicle and transporting the maximum rated payload over level, smooth, improved, and prepared hard-surface roads at a sustained speed of 50 miles per hour (mph), shall track towing vehicle without swerving from side to side over 3 in. to each side of the path of the towing vehicle when it is moving in a straight line (see 4.6.18).

3.5.5.1 Cross-country operation. The semitrailer, when transporting the rated payload over unimproved roads, trails, open fields, rolling hills, and other cross-country terrain at speeds up to 15 mph, shall operate without failure, permanent deformation, or damage to itself or to the towing vehicle (see 4.6.19).

3.5.6 Gradeability. When towed over a dry, hard surface free of loose material, the semitrailer shall exhibit gradeability and stability specified herein (see 4.6.19).

3.5.6.1 Longitudinal grades. The semitrailer shall follow the towing vehicle without weaving to the extent affecting the controllability of the vehicle combination when ascending or descending longitudinal grades having a 20 percent grade (see 4.6.19).

3.5.6.2 Side slopes. The left and right sides of the semitrailer shall follow the towing vehicle without slipping or upsetting when operating on 20 percent side slopes (see 4.6.19).

3.5.7 Electromagnetic interference. When specified (see 6.2), the complete vehicle shall be shielded against electromagnetic interference in accordance with the tactical vehicle requirements of Class IIIA of MIL-STD-461 and MIL-STD-285. The frequency range for the shielding and attenuation in decibels shall be specified by the user (see 4.6.20).

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3.6 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2), all cleaning, treatment, and painting shall be in accordance with the manufacturer's standard practice. Color required shall match color chips of FED-STD-595.

3.7 Marking. Vehicle marking shall be in accordance with MIL-STD-642.

3.8 Data plates. All data plates shall be in accordance with the manufacturer's standard practice.

3.9 Workmanship. The workmanship displayed in the fabrication and assembly of the semitrailers shall be such as to meet the performance under all applicable environmental conditions. The quality of workmanship shall assure delivery of semitrailers, which are free from defects, improper manufacturing or assembly practices, and which meet or exceed requirements specified herein. Defective components or parts and assemblies, which have been repaired or modified to overcome deficiencies, shall not be furnished. Painted surfaces shall be free from sags or runs.

3.10 Air transportability. If specified, the design for air transportability shall conform to MIL-STD-1791.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on the first production semitrailer and on two additional semitrailers at the initial production. This inspection shall include the examinations of 4.4 and the tests of 4.5.

4.2.1 First production vehicle inspection. The first completed production vehicle shall be subject to provisional inspection by the Government, after a complete inspection by the contractor which includes road test in accordance with 4.2. 1. 1.

4.2.1.1 Fifty mile test. The semitrailer shall be loaded with highway payload or with a simulated load of equal weight and operated for a distance of not less than 50 miles. Test course selected by the manufacturer shall be on smooth, approximately level, hard-surfaced roads.

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4.2.2 Initial production vehicle test. After completion of preliminary inspection of the first semitrailer, the second and third produced shall be subjected to examination and tests specified herein and the 8000 mile test specified in table II.

TABLE II. 8000 mile test.

Course	Mileage and speeds
High speed paved road (APG Munson Test Area)	40% up to 50 mph maximum
Improved gravel road (APG Munson Test Area)	27% up to 25 mph maximum
Cross-country terrain (Perryman Course)	25% up to 15 mph maximum
No. 1 secondary roads (APG Perryman Test Area)	
Hilly secondary road (Churchville Test Course C)	6% up to 15 mph maximum
Belgian block (APG Munson Test Area)	2% up to 15 mph maximum

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests in accordance with table IV.

4.3.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

4.4 Examinations. Visual, dimensional, and primary functional examination shall be performed on each semitrailer for conformance to the technical data package and this specification. A minimum of examination shall be performed against the classification of defects shown in table III. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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TABLE III. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Frame; structural, welding, riveting, or sheet metal defects (see 3.4).	Visual
102	Fifth wheel plate and kingpin; improper mounting and assembly; welding of locking action (see 3.4).	Visual
103	Axles; improper assembly and mounting; welding defects (see 3.4).	Visual
104	Wheels and tires; tire damage (see 3.4.1.1).	Visual
105	Electrical system components, and circuits; malfunction; inoperative, damage (see 3.4.2).	Visual and functional
106	Doors; improper fits, water leaks (see 3.4.5).	Visual and functional
107	Tiedown devices; improper mounting and assembly welding defects (see 3.4.9).	Visual and functional
108	Landing gear; malfunction extended, retreated stowed or locking action (see 3.4.10).	Visual and functional
109	Gaskets and seals, improper fits, leaks (see 3.4.11).	Visual and functional
110	Trailing ability- nonconformance (see 3.5.5).	Visual and functional
111	Turning ability; nonconformance (see 3.5.1).	Visual and functional
112	Service brakes; malfunction; leak; nonconformance to stopping distance (see 3.5.2).	Visual and functional
113	Towing speeds, highway; nonconformance (see 3.5.2).	Functional
114	Automatic brake actuation; malfunction; nonconformance (see 3.5.2.1).	Functional
Minor:		
201	Wheels and tires; improper size, type, and mounting (see 3.4.1.1).	Visual
202	Wiring or tubing; improperly mounted; coded; or protected; defective (see 3.4.2.2).	Visual

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TABLE III. Classification of defects - Continued.

Category	Defect	Method of examination
203	Electrical system components; improper installation or assembly (see 3.4.2.2).	Visual
204	Landing gear; improper assembly or installation (see 3.4.10).	Visual

4.4.1 Unclassified defects. All component defects having no bearing on function, safety, interchangeability, or life, but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or in two lots or more within a thirty-day period, will be added to the minor classification.

4.4.1.1 Recurring major deficiencies. A major deficiency (see 6.4) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4.1.2 Recurring minor deficiencies. A minor deficiency (see 6.4) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

4.5 Acceptance tests. Each semitrailer shall be operated for a distance of not less than five miles by the contractor. Semitrailers shall be completely assembled and serviced. Acceptance test shall include all acceptance tests specified at the place of manufacture and indicated in table IV. Performance of semitrailers shall be demonstrated on smooth, relatively level, hard-surfaced roads and shall meet all requirements specified herein. After completion of the five mile road test, doors shall be opened and closed to assure proper operation, fit, alignment, and action of control devices.

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TABLE IV. Classification test.

Title	Require-ment	Test	Place of manufacturer	Government proving ground
Spare wheel assembly check	3.4.1	4.6.3	X	X
Electrical system check	3.4.2	4.6.4	X	X
Body air leakage test	3.4.3	4.6.5	X	X
Waterproofness test	3.4.4	4.6.6	X	X
Adjustment mechanisms check	3.4.5	4.6.7	X	X
Controls test	3.4.6	4.6.8	X	X
Air hydraulic lines, and fittings check	3.4.7	4.6.9	X	X
Lifting attachments and tiedown devices check	3.4.8 & 3.4.9	4.6.10	X	X
Landing gear check	3.4.10	4.6.11	X	X
Seals check	3.4.11	4.6.12	X	X
Turning ability test	3.5.1	4.6.14	X	X
Service brakes and automatic actuation device tests	3.5.2	4.6.15	X	X
Fording test	3.5.3	4.6.16	X	X
Environmental test	3.5.4	4.6.17	--	X
Towing speeds and trailing ability test	3.5.5	4.6.18	--	X
Cross-country and gradeability test	3.5.5.1 thru 3.5.6.2	4.6.19	--	X
Electromagnetic compatibility test	3.5.7	4.6.20	--	X

4.6 Method of inspection.

4.6.1 Materials and construction. Conformance to 3.2 and 3.4, inspection and material welding certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.4, 3.6 and 3.7, shall be determined by examination for the defects listed in table III. Examination shall be visible, tactile or by measurement with standard inspection equipment.

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4.6.3 Spare wheel assembly check. To determine conformance to 3.4.1 through 3.4.1.2, each assembly shall be checked for proper tire size and assembly.

4.6.4 Electrical system check. To determine conformance to 3.4.2 through 3.4.2.2, the electrical system shall be checked for transmitting proper voltage, free of shorts, and have continuity from end to end of all circuits.

4.6.5 Body air leakage test. To determine conformance to 3.4.3, each semitrailer van body shall be subjected to an air leakage test by closing all doors and openings and forcing air at specified ft^3/min from the outside to the inside of the van body. The air pressure shall be not less than that specified.

4.6.6 Waterproofness test. To determine conformance to 3.4.4, the semitrailer shall be subjected to a spray test simulating the specified rainfall as measured on the surface of the van body by a U.S. Weather Bureau type gauge. The simulated rainfall shall be produced by means of water spray nozzles of such design that the water is emitted at a distance of not more than two feet from the sprayed surface in the form of droplets having the specified size and angle. The water supply pressure shall be adequate for simulating the specified rainfall. Doors and other openings in the van body shall be closed during the test. Openings required to be open during operation of the interior equipment and provided with covers preventing rain under a maximum angle of 45 degrees from the vertical shall be open during the test. After the test, the van interior shall be inspected for water penetration and point of entry. To facilitate checking for leaks, no water panels, insulation and sound reducing material shall be installed prior to the test except for the insulation between the outer skin and the fording plate, and in the door. The electrical system shall be energized, external outlets inspected, and air conditioners and similar equipment (if installed) shall be operated for a time period sufficient to determine proper functioning. Should leaks or defects be noted, they shall be repaired or reported, respectively, and the test repeated. The contractor's spray facility must be approved by the U.S. Army Tank-Automotive Materiel Readiness Command to guarantee compliance with the specified requirements.

4.6.7 Adjustment mechanisms check. To determine conformance to 3.4.5, all electrical, mechanical and hydraulic adjustment mechanisms, shall be checked for proper adjustment and adjusted, if required.

4.6.8 Controls check. To determine conformance to 3.4.6, all controls shall be operated and checked for functional requirements.

4.6.9 Air hydraulic lines and fittings check. To determine conformance to 3.4.7, all lines and fittings shall be checked for proper installation and leaks.

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4.6.10 Lifting attachment and tiedown devices check. To determine conformance to 3.4.8 and 3.4.9, each semitrailer shall be checked for proper location and installation of lifting attachments and tiedown devices.

4.6.11 Landing gear check. To determine conformance to 3.4.11, the landing gear shall be functionally operated and checked for completeness of assembly and for proper mounting.

4.6.12 Seals check. To determine conformance to 3.4.11, after the van body fording test, the bearings shall be checked for water contamination and lubricant leakage.

4.6.13 Towing speeds and trailing ability check. To determine conformance to 3.5 and 3.5.5, each semitrailer without load shall be coupled to towing vehicle, operated as specified, and checked for trailing ability at all towing speeds.

4.6.14 Turning ability test. To determine conformance to 3.5.1, each semitrailer, when coupled to towing vehicle, shall be operated as specified and checked for turning ability.

4.6.15 Service brakes and automatic actuation device test. To determine conformance to 3.5.2 and 3.5.2.1, each semitrailer, when coupled to towing vehicle without load, shall be operated as specified and checked for functional requirements of service break and automatic actuation device.

4.6.16 Fording test. To determine conformance to 3.5.3, the semitrailer shall be immersed into or towed in water at the depth and for the time period as specified. During the test, openings above the water level shall be open. To facilitate inspection after the test, the vehicle shall be empty, no floor covering installed, however, the insulation between the outer skin and the fording plate shall be in place. After the test, all the components submerged during fording shall be inspected, the electrical system energized, and the vehicle towed a distance sufficient to determine proper functioning of the brakes and other equipment, such as generators, operated for a time period sufficient to determine their proper operation. Should leaks or defects be noted, they shall be repaired or reported, respectively, and the test repeated.

4.6.17 Environmental test. To determine conformance to 3.5.4, the vehicle shall be tested for meeting the environmental requirements.

4.6.18 Towing speeds and trailing ability test. To determine conformance to 3.5.5, the combination operated, as specified, shall be tested and checked for meeting performance requirements.

4.6.19 Cross-country and gradeability tests. To determine conformance to 3.5.5.1 through 3.5.6.2, the combination operated, as specified, shall be tested and checked for meeting performance requirements.

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4.6.20 Electromagnetic interference test. To determine conformance to 3.5.7, the vehicle shall be tested for meeting compatibility requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Semitrailers covered by this specification are intended for use by the U.S. military services as vehicles suitable or adaptable for housing and transporting delicate electronic and other similar equipment requiring a relatively soft ride and maximum protection during military operations over all types of roads, fording of water crossings, and extremes of temperature and climate. For surface transportation, the semitrailers are to be towed by the truck tractor, 5 ton, 6x6, M52, or M818.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Classification of semitrailer required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
d. If first article samples are required (see 3.1).
e. If tires and inner tubes are other than as specified (see 3.4.1.1 and 3.4.1.2).
f. Suppression of electromagnetic interference, as required (see 3.5.7).
g. Applicable requirements for treatment and painting of vehicles (see 3.6).
h. If first article inspection is other than as specified (see 4.2).
i. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Electronic equipment transporter
Housing electronic equipment
Towing

6.4 Definitions of recurring major and minor defects. The Government inspector will verify that thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but, also, for any other departures from good workmanship. The Government inspector will assure that all deficiencies encountered during the inspection are enumerated on the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in ANSI/ASQC Z1.4. Corrective action will be taken for recurring deficiencies.

6.5 Recycled materials. The use of recycled materials, which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item, will be encouraged.

6.6 Recommended service products. Typically, the semitrailer should be serviced with the following product listed in table V or equivalent.

TABLE V. Service product specifications.

Product use	Ambient air temperature operating range	
	0° to +125°F	-50° to 0°F
Fluid:		
For hydraulic brakes	MIL-B-46176	MIL-B-46176
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924
Oil:		
For general purpose lubrication	MIL-L-3150	VV-L-800

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-46756E(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 970415

3. DOCUMENT TITLE
SEMITRAILER, VAN: 10 TON, 4-WHEEL SERIES

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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